

Phuong-Linh Huynh-Ha

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RESEARCH EXPERIENCE

- **KIT Computer Vision Lab – Kyoto Institute of Technology (KIT)** Jan. 2026 – Mar. 2026
Research Intern, supervised by Prof. Nobuhara Shohei
 - Awarded the **JASSO Scholarship by the Japanese government** and selected for the **KIT Global Internship Program** based on academic excellence.
 - Developing a **3D reconstruction framework** for **near-rotational motion** and **low-parallax environments**. Resolved motion ambiguity in **feature-poor scenes** by integrating **2D correspondences** to rectify and enhance SLAM-based estimations.
- **Software Engineering Laboratory – SELab HCMUS** Aug. 2025 – Present
Thesis Student - Supervised by Assoc. Prof Minh-Triet Tran on Computer Vision
 - Co-authored and published "**Generalizability Evaluation and Anchor-Guided Approach for Category-Agnostic Pose Estimation**" at SOICT 2025 (Springer CCIS).
 - The thesis focuses on developing a method called "**EndoFGS**" that works on the **frequency domain** for **surgical scene reconstruction** designed to overcome challenges in feature-poor environments, such as blood vessels and tissue surfaces.
 - Participating in the **CVPR 2026 Spiideo SoccerNet SynLoc Challenge**; developing methods to detect and localize athletes in single-frame world coordinates utilizing synthetic data.

SELECTED PUBLICATIONS

- Hoang-Phuc Nguyen, Phuong-Linh Huynh-Ha, Minh-Triet Tran. **Generalizability Evaluation and Anchor-Guided Approach for Category-Agnostic Pose Estimation**. *SOICT 2025, Springer CCIS*.
Introduced a **training-free, anchor-guided geometric mapping approach** using Delaunay triangulation to resolve overfitting and improve fine-grained keypoint accuracy in category-agnostic pose estimation.

EDUCATION

- **University of Science (HCMUS), Ho Chi Minh City, Vietnam** Sep. 2022 – Present
Advanced Program in Computer Science; GPA: 3.9/4.0
 - Granted **Merit-Based Scholarship (Top 4 GPA)** in Semester 2, 2024-2025.
 - Being 1 of **5 recipients** of **Career Readiness Scholarship**, awarded by **Gameloft** in partnership with **HCMUS** for demonstrating strong professional potential and academic merit.
 - Achieved **Top 1.9% score on the VNU-HCMC Assessment Test (HSA)**
 - Relevant coursework: Artificial Intelligence, Computer Vision, Natural Language Processing, Information Retrieval, Computer Systems, Statistics.

ACHIEVEMENTS AND AWARDS

- **Artificial Intelligence Olympiad for University Students (3-person team)** Nov. 2025 – Dec. 2025
Organized by Vietnam Association for Information Processing (VAIP)
 - Awarded **Third Prize** at the **National Finals** (Dec. 2025).
 - Awarded **First Prize, Champion** across all **Regional Preliminary Rounds** (Nov. 2025).
- **Odon Vallet Scholarship** Aug. 2022
 - Awarded prestigious **Odon Vallet Fellowship** for academic excellence, issued by Rencontres du Vietnam.
- **Vietnam National Olympiad in English** Apr. 2022, Feb. 2021
 - **Bronze Medal** in 2022 and **Honorable Mention** in 2021, awarded by Vietnam Ministry of Education and Training.
 - **First Place** in the Provincial Olympiad in English of Quang Ngai throughout 2021 - 2022.

PROJECTS

- **Robust 3D Reconstruction via 2D Correspondence Integration** Researching
 - Extracting and comparing optical flow from MegaSaM and CoTracker3 to resolve motion ambiguity in low-parallax and near-rotational videos.
 - Using heatmap analysis to visualize flow differences, creating a reliable method to correct SLAM tracking errors in feature-poor environments.
- **EndoFGS: Surgical Scene Reconstruction & 3D Gaussian Splatting** Researching
 - Building a surgical scene reconstruction pipeline using 3D Gaussian Splatting, evaluated on the EndoNeRF and StereoMIS datasets.
 - Improving rendering quality (PSNR/SSIM) on textureless tissues by applying frequency-domain filters, including Wavelet, Fourier, and high-boost, to enhance image features.
- **SoccerNet SynLoc: Athlete Detection & Localization** Researching
 - Developing a monocular athlete localization pipeline for the CVPR 2026 challenge using YOLO26 for bounding box detection and RTMPose for pelvis keypoint extraction, which then needs to be projected down to stadium surface.
 - Mapping 2D pixel coordinates to 3D stadium coordinates by applying pitch-based camera calibration and ground projection techniques.

EXTRACURRICULAR ACTIVITIES

- **Youth Union Branch Commendation** 2022–2026
 - *University of Science, VNU-HCM*
 - Awarded **Certificate of Merit** from the Principal for achieving good results in Youth Union organizational work and youth movements during the 2022-2023 academic year, in the board of Committee Member of Youth Union Branch 22APCS2).
 - Awarded **Title of Outstanding Youth for Following President Ho Chi Minh's Teachings** in March 2026.
- **Inspire to Aspire Seminar Series** 2023
 - *Organizer – Ho Chi Minh City, Vietnam*
 - Organized a seminar series titled “*Inspire to Aspire*” that connected students with experienced professionals in technology and business, providing career insights and fostering industry awareness.
- **Asian-Pacific Children's Convention (APCC)** Jul. 2015
 - *Youth Delegate – Fukuoka, Japan*
 - Selected by the Ho Chi Minh Communist Youth Union to represent Vietnam at APCC in Fukuoka, Japan, focusing on international cultural exchange to foster global understanding, peace, and friendship.

SKILLS & INTERESTS

Technical Skills:	Scientific Research, Problem Solving, Machine Learning, Computer Vision
Technical Languages:	C/C++, Python, Kotlin, SQL, LaTeX, Markdown, CSS, HTML
Frameworks/Libraries:	PyTorch, NumPy, Pandas, R, Jetpack Compose, SFML
Tools:	Google Colab, Docker, Git/GitHub, Android SDK, 3D Slicer
Languages:	English (IELTS 8.0), Vietnamese (native), Japanese (for daily conversation)
Interests:	Cycling, Swimming, Singing, Guitar (learning), Poker